Pavan Kumar Balija

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PROFILE OVERVIEW

- Software Developer with **4** years of experience in the **Modern C++ (C++11/14/17) development** with **Adaptive AUTOSAR platform.**
- Proficient in designing and developing **Basic Software Components** in compliance with **Adaptive AUTOSAR specifications**.
- Successfully **designed and implemented sample applications** to validate AUTOSAR functionalities and features.
- Hands-on experience with **Adaptive AUTOSAR configuration tools** and integration workflows.
- Involved in the **development and functional testing** of key Adaptive AUTOSAR modules such as **Cryptography**, **ara::core**, **Diagnostics**.
- Strong technical knowledge of Adaptive AUTOSAR components, including **COM, SM (State Management), DIAG (Diagnostics), CRYPTO(Cryptography), and EXEC (Execution Manager), etc...**
- Experience in **integrating the Adaptive AUTOSAR** stack on **QEMU virtual platforms**, ensuring end-to-end stack functionality and validation.
- Well-versed in **software development methodologies** including **Agile, V-Model (SDLC)**, and coding standards such as **AUTOSAR C++14 (MISRA compliant).**
- Managing **customer queries**, involving in issues and **bug fix of the products**.
- Experience in working with continuous integration and continuous deployment (CI/CD) environment **using Jenkins, Jira, Bitbucket, Git, Jfrog Artifactory, Confluence, Docker, VS Code, etc…**
- Demonstrated ability to **analyse customer requirements** and translate them into functional, modular, and scalable software components.
- A **proactive team player** with strong **analytical, problem-solving**, and **interpersonal skills**, committed to delivering high-quality embedded solutions in automotive domains.

TECHNICAL SKILLS:

Programming Languages:	Modern C++, C++ (11/14/17), STL, Multithreading and C - Language.
Development Tools	Visual Studio Code, CANoe, GitHub, Vector Cast Tool, Davinci developer Adaptive tool, Atlassian Tools (Bitbucket, Jira), Code Beamer (ALM), Axivion tool, AWS EC2 Instance, Shell Scripting, CMake, JSON, Yocto, QEMU.
Operating System	Linux (Ubuntu), Windows, QNX.
Domain Expertise	Adaptive AUTOSAR, Classic AUTOSAR and ADAS.
Software Methodologies	Agile Methodology, V-Model (SDLC), AUTOSAR C++14 (MISRA compliant)
Protocols & System Programming	TCP/IP, UDP, SOME/IP, IPC Mechanisms.

EDUCATION:

Qualification	Institution	Board/University	Passing year	Marks In %
B.Tech (ECE)	SMEISGI	JNTUA	2020	70.68
Class XII	GOVERNMENT JUNIOR COLLEGE	BOARD OF INTERMEDIATE, AP	2016	87.9
Class X	S R K MPL BOYES HIGH SCHOOL	BOARD OF SECONDARY, AP	2014	80

PROFESSIONAL EXPERIENCE:

- **KPIT Technologies Ltd** (Oct/24/2024 – Present) – Sr Software Engineer – **Modern C++, Adaptive AUTOSAR**
- **AVIN Systems Pvt Ltd** (July/15/2021 – Oct/23/2024) – Sr Software Engineer – **Modern C++, Adaptive AUTOSAR**

AWARDS:

- **Thankyou Award**, AVIN Systems Pvt Ltd, Dec 2023.
  Description: Completing activities with minimal guidance.
- **Excellence Award**, AVIN Systems Pvt Ltd, Jan 2024.
  Description: Having an ability to maneuver through multiple domains with commitment & dedication.
- **Above & Beyonders Award**, AVIN Systems Pvt Ltd, Jan 2024.
  Description: In recognition and appreciation of outstanding performance and contribution to the project.

PROJECTS UNDERTAKEN:
PROJECT 1:

Project Name	**Integration and Development of Diagnostic Stack**
Project Description	Bug fixing and Implementation of overall **DIAGNOSTIC** according to AUTOSAR specification and customer requirements.
Contribution	**My responsibilities included:** ➢ Analysed customer requirements based on **AUTOSAR Diagnostic specification doc**. ➢ Identified and **fixed all defects** in the Diagnostic stack and developed code as per updated requirement. ➢ Performed **unit testing** of the complete **Diagnostic stack** using the **VectorCAST tool**. ➢ Resolved all **static code violations in the Diagnostic stack**, ensuring compliance with coding standards. ➢ Participated in **ASPICE activities**, contributing to process improvement and quality assurance.
Tools Used	Visual Studio Code, Ubuntu, JIRA, CANoe, GitHub, Vector Cast Tool, Davinci tools.

PROJECT 2:

ProjectName	**ara::core Module Development on Adaptive AUTOSAR Platform**
Project Description	Implementation and integration of overall Adaptive AUTOSAR FC's according to AUTOSAR specification and customer requirements. **Adaptive ara::core** is one of the module used to provide the advance C++ libraries to all AUTOSAR modules.
Contribution	**My responsibilities included:** ➢ Analysed requirements from the **AUTOSAR Adaptive ara::core SWS (20-11)** specification doc. ➢ Integrated the **ICEORXY C++ feature** into the **ara::core module** and modified code based on requirements. ➢ Performed **unit testing** of the ara::core module in alignment with development code. ➢ Conducted **integration testing** to validate overall module functionality and component interactions. ➢ Created **sequence diagram scripts** and generated diagrams using **PlantUML** for architectural visualization. ➢ Resolved **static code violations** in the Adaptive Core codebase to maintain coding standard compliance.
Tools Used	Visual Studio Code, Atlassian Tools (Bitbucket, Jira), Code Beamer (ALM), GitHub, Axivion tool, JIRA.

PROJECT 3:

ProjectName	**Cryptography Module Development on Adaptive AUTOSAR Platform**
Project Description	Implementation and integration of overall Adaptive AUTOSAR FC's according to AUTOSAR specification and customer requirement. **Cryptography** is one of the modules used to provide **cybersecurity** of the applications in the vehicle from the hacker attacks and to verify the authenticity and integrity of the data.
Contribution	**My responsibilities included:** ➢ Analysed requirements from the **AUTOSAR Cryptography SWS (20-11)** specification document. ➢ Implemented **Cryptography Functional Cluster (FC)** based on the specifications defined in the **AUTOSAR_SWS_Cryptography** document. ➢ Configured **Cryptography interfaces** using the **ASTRA configuration tool (**Davinci developer Adaptive). ➢ Developed **basic test applications** (such as **qt-app** and **qt-tester**) to invoke and validate **Cryptography APIs**. ➢ Configure the **SOME/IP** (P-port and R-port) services on Davinci developer Adaptive. ➢ Performed **functional testing** of the Cryptography module in line with the development code. ➢ Created **sequence diagram scripts** and generated diagrams using **PlantUML** for system behaviour visualization. ➢ Resolved **static code violations** in the Cryptography development code.
Tools Used	VS Code, Davinci developer Adaptive tool, Bitbucket, Jira, Code Beamer, GitHub, Axivion tool, CANoe.

PROJECT 4(Current):

ProjectName	**Integration and Validation of Adaptive AUTOSAR Stack on Virtual Platforms (QEMU)**
Project Description	Involved in Integration the Adaptive AUTOSAR stack on **QEMU** environments. The project required ensuring that the stack worked as expected in these environments, identifying any issues, providing solutions, and documenting the changes encountered. This also included validation of the stack's performance, functionality, and integration with the virtual platforms.
Contribution	**My responsibilities included:** ➢ Integrated the **Adaptive AUTOSAR stack** into **QEMU virtual environments**. ➢ Ensured proper **functionality and performance** of the stack within virtualized platforms. ➢ Identified defects, **troubleshoot issues**, and implemented effective solutions. ➢ Utilized **Yocto** to build the Adaptive AUTOSAR stack and generate the required **ext4** file and **kernel image**. ➢ Deployed the generated images into the **QEMU virtual environment** for validation and testing. ➢ **Developed communication** between the **QEMU platform** and a **Linux platform** to **send and receive data using the SOME/IP protocol**, including code implementation and modifications as per requirements. ➢ **Documented challenges** encountered and detailed the solutions applied during the integration and testing phases.
Tools Used	Visual Studio Code, AWS EC2 Instance, Ubuntu, Yocto, QEMU.

STRENGTHS:

- Positive Learning Attitude.
- Dedication and Flexible to work.
- Team Worker.
- Good listener and communicator.
- Quick Learner.

DECLARATION:

I do hereby declare that the particulars of information and facts stated here in above are true, correct and complete to the best of knowledge and belief.

Yours sincerely
Balija Pavan Kumar